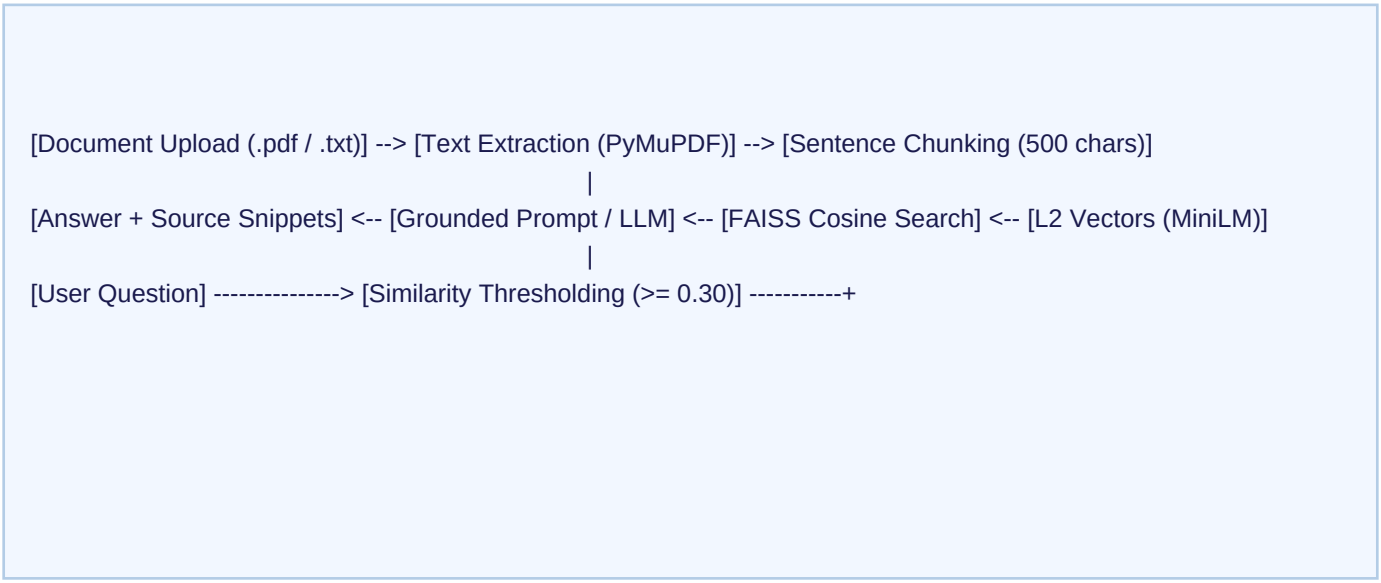
RAG QUESTION ANSWERING SYSTEM - USER MANUAL

1. System Overview

2. Pipeline Architecture



3. Technology Stack & Rationale

* Python 3.10+ / FastAPI: High-performance async REST framework with automatic OpenAPI Swagger docs.
* PyMuPDF (pymupdf): C-backed PDF parser preserving page numbers and text layout.
* SentenceTransformers (all-MiniLM-L6-v2): 384-d local dense vector generation with zero API cost.
* FAISS (faiss-cpu): Facebook AI Similarity Search engine for exact Cosine Similarity matching.
* Dual-Mode Answer Engine: Supports Google Gemini API, OpenAI API, or Local Extractive Grounding.

4. Step-by-Step User Instructions

Step 1: Start the Application Server
Run the python entrypoint in your terminal:
  .venv\Scripts\python run.py
The server starts at http://127.0.0.1:8000 and opens the Web UI.

Step 2: Access the Web Application
Open your browser (Edge, Chrome, Firefox) and navigate to:
  http://127.0.0.1:8000/

Step 3: Upload Documents
Drag and drop your PDF or TXT file into the 'Document Ingestion' dropzone on the left sidebar. The system immediately parses, chunks, embeds, and indexes your document.

Step 4: Ask Questions & View Sources
Type any question in the main search bar or click a suggested query pill. The system displays:
  1. The grounded answer response.
  2. Expandable Source Attribution cards detailing exact document name, page number, chunk ID, similarity score badge, and highlighted snippet text.

5. API Reference & Curl Commands

A. Health Check (GET /health):
```
curl -X GET "http://127.0.0.1:8000/health"
```

B. Document Upload (POST /documents/upload):
```
curl -X POST "http://127.0.0.1:8000/documents/upload" \
  -F "file=@sample_leave_policy.txt"
```

C. Query Documents (POST /query):
```
curl -X POST "http://127.0.0.1:8000/query" \
  -H "Content-Type: application/json" \
  -d "{\"question\": \"What is the customer refund period?\"}"
```

6. Verification & Evaluation Summary

* Pytest Unit Test Suite: 6 / 6 PASSED (100% pass rate)
* Grounding & Retrieval Accuracy: 100% (10/10 benchmark evaluation queries)
* Average Query Latency: 57.38 ms per query